DropWise — visual annex

Figure 1. How the experiment is built

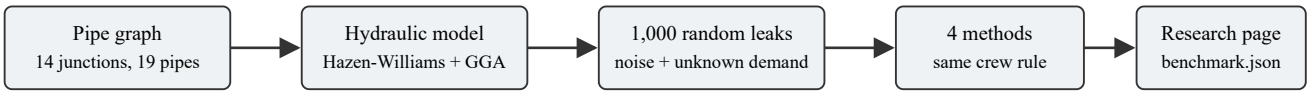
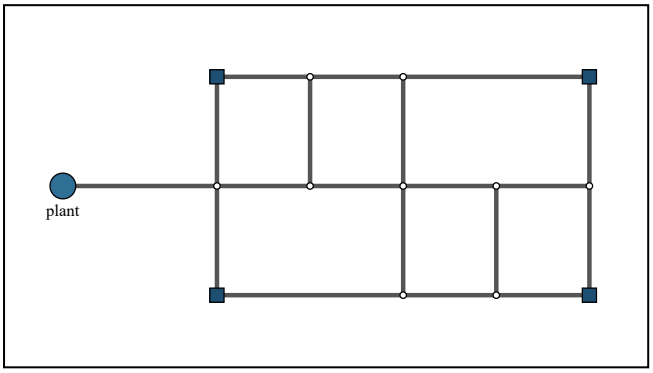
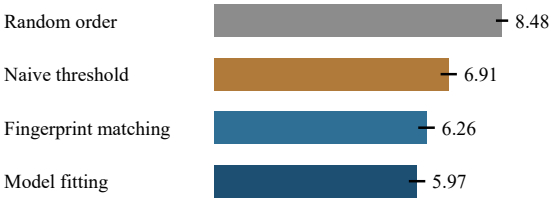


Figure 2. The network and its four sensors



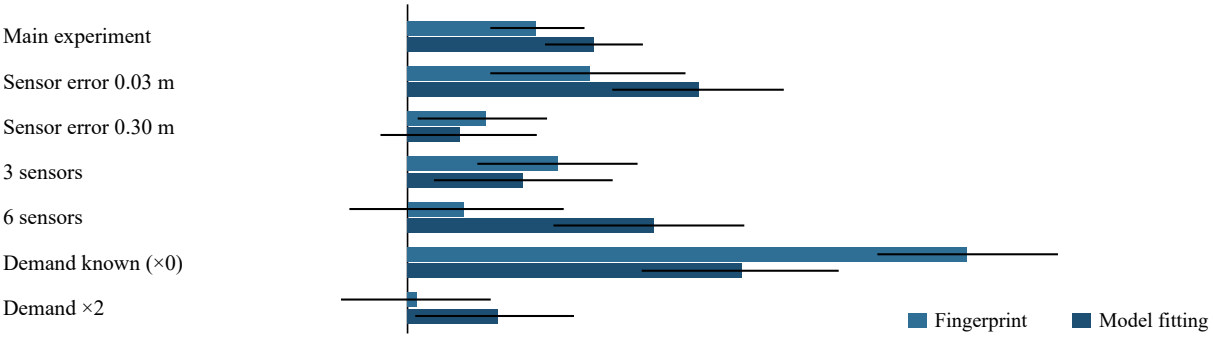
Squares: pressure sensors. Circles: junctions. 13.4 km of pipe.

Figure 3. Mean dry holes (95% CI), 1,000 leaks



Fewer is better; 10 = not found. Pre-registered targets: H1 partially supported, H2 partially supported.

Figure 4. Dry holes saved against the naive rule, by condition (95% CI)



The saving depends most on how well demand is known. Each condition uses 300 leaks; the main experiment uses 1,000.

Table 1. Cost and payback (assumptions on the Research page)

Item	Euro	Lei / note
Dry hole (dig + water lost while it takes place)	€980	19,781 lei
Saved per leak: fingerprints / model fitting, today	€635 / €921	12,818 lei / 18,594 lei
Payback of a €12,000 system, today	138 months / 66 months	4 large leaks a year
Payback if demand were known	16 months / 31 months	smart meters needed
Water lost by one leak, per day	1,037 m³	14,546 lei at 14.03 lei/m³

Rate €1 = 20.18 lei. Equipment prices: Industrial pressure transmitter, 0-10 bar: EUR 165.99 and EUR 215 · Pressure data logger with modem: USD 1,095 to 1,925 (currency not stated on the page) · Commercial acoustic leak-noise logger kits: USD 13,500 for a standard system, USD 22,500 for a hybrid system · Satellite radar leak survey: USD 85,500 for 181 miles of pipe (about USD 470 a mile, USD 290 a km). They were seen in web searches and several pages could not be opened.